Flow Matching for Generative Modeling

Yaron Lipman[1,2] **Ricky T. Q. Chen**[1] **Heli Ben-Hamu**[2] **Maximilian Nickel**[1] **Matt Le**[1]
[1]Meta AI (FAIR) [2]Weizmann Institute of Science

(Lipman et al., ICLR 2023)

이소윤 | synuolee@gmail.com

Contents

0. Abstract
1. Introduction
2. Preliminaries: Continuous Normalizing Flows
3. Flow Matching
4. Conditional Probability Paths and Vector Field
5. Related Work
6. Experiments
7. Conclusion

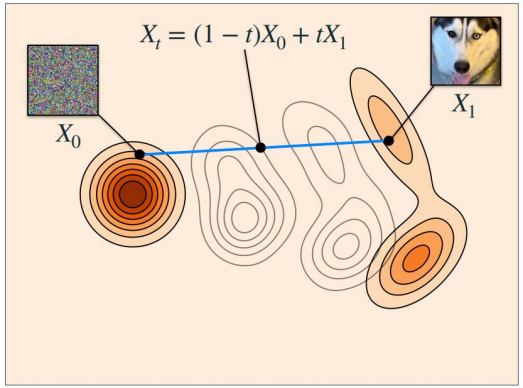
0. Abstract

Flow Matching (FM)

- a **simulation-free** approach for training **CNFs**
- based on **regressing vector fields** of fixed conditional probability path

- employ FM with non-diffusion path by **Optimal Transport (OT)** displacement
- faster training and sampling, better generalization than diffusion paths

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}\left\| u_t^\theta(X_t) - (X_1 - X_0) \right\|^2$$



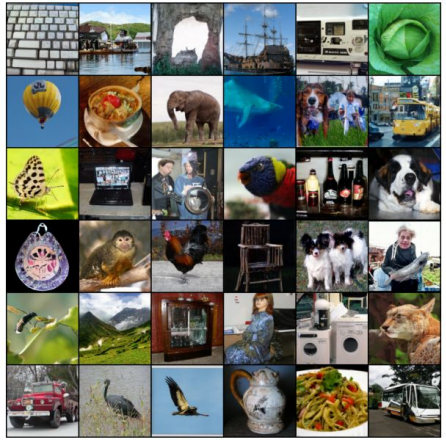


Figure 1: Unconditional ImageNet-128 samples of a CNF trained using Flow Matching with Optimal Transport probability paths.

1. Introduction

Flow based Model

확산 모델의 성공과 한계:

- 확산 모델 (DDPM, Score-based models)이 이미지 생성에서 큰 성공
- 단순한 확산 과정에 국한 → 제한된 확률 경로 공간
- 매우 긴 훈련 시간, 비효율적 샘플링

CNF의 가능성과 기존 한계:

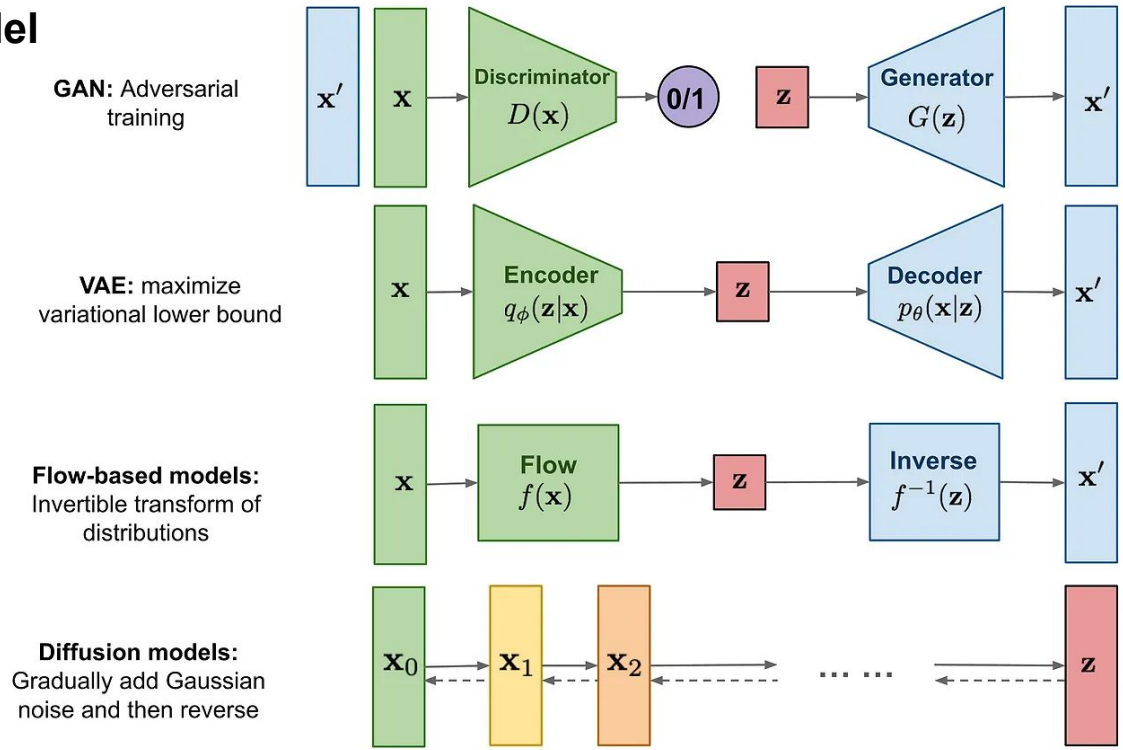
- CNF는 임의의 확률 경로를 모델링 가능 → 확산 모델보다 일반적
- 최대 우도 훈련: 비싼 ODE 시뮬레이션 필요
- 기존 시뮬레이션-프리 방법: 다루기 어려운 적분(Rozen et al.) 또는 편향된 그래디언트(Ben-Hamu et al.)

FM의 목표:

- 시뮬레이션 없이 CNF를 효율적으로 훈련
- 확산을 넘어선 일반적 확률 경로 사용 가능
- 확산 과정 없이 확률 경로를 직접 다루기

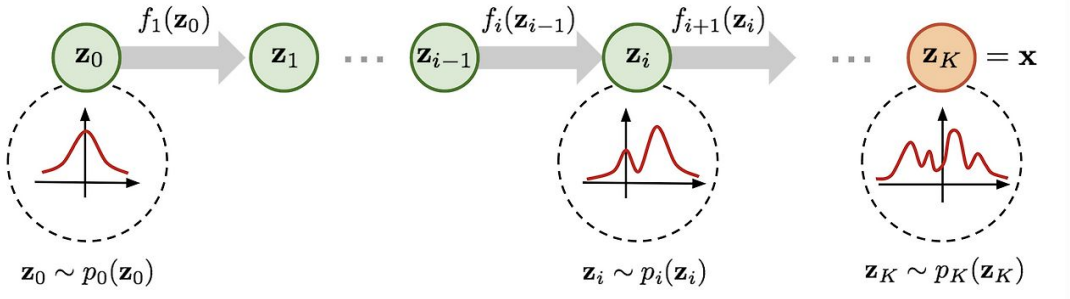
1. Introduction

Flow based Model

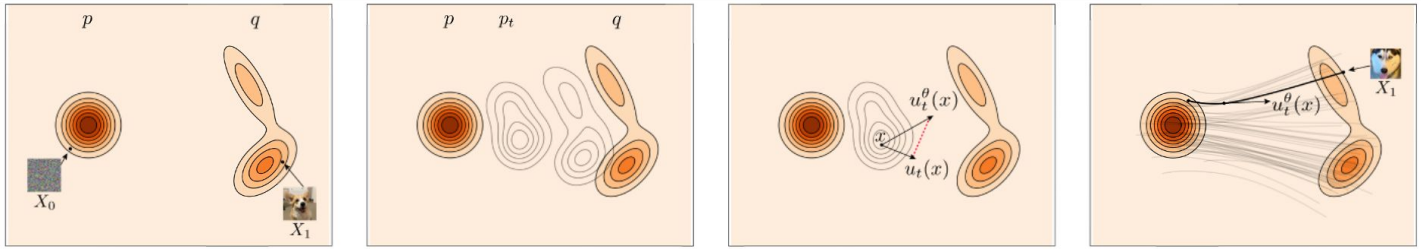


1. Introduction

Normalizing Flows (NF)



Continuous Normalizing Flows (CNFs)



(a) Data. **(b)** Path design. **(c)** Training. **(d)** Sampling.

2. Preliminaries: Continuous Normalizing Flows (CNFs)

$\mathbb{R}^d$ denote the data space with data points $x = (x^1, \ldots, x^d) \in \mathbb{R}^d$.

Two Important Objects

1. time-dependent **probability density path** $p : [0,1] \times \mathbb{R}^d \to \mathbb{R}_{>0}$
2. time-dependent **vector field** $v : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$.

Flow $\phi : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$

- vector field construct a time-dependent diffeomorphic map, called a *flow*

$$\frac{d}{dt}\phi_t(x) = v_t(\phi_t(x))$$

$$\phi_0(x) = x$$

2. Preliminaries: Continuous Normalizing Flows (CNFs)

Continuous Normalizing Flow (CNF)

- modeling the vector field v_t with a neural network $v_t(x;\theta)$
  - $\theta \in \mathbb{R}^p$ are its learnable parameters
- reshape a simple p0 to a more complicated p1
  - via the push-forward equation $p_t = [\phi_t]_* p_0$
  - push-forward operator $*$ defined : $[\phi_t]_* p_0(x) = p_0(\phi_t^{-1}(x)) \det \left[\frac{\partial \phi_t^{-1}}{\partial x}(x) \right]$

3. Flow Matching

Flow Matching (FM) objective

- Given a target probability density path $p_t(x)$ and a corresponding vector field $u_t(x)$

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{t,p_t(x)} \|v_t(x) - u_t(x)\|^2$$

where $t \sim \mathcal{U}[0,1]$, and $x \sim p_t(x)$

- θ : the learnable parameters of the CNF vector field v_t
- the FM loss **regresses the vector field** u_t with a neural network v_t
- the learned CNF model will generate $p_t(x)$

Issue

- **intractable** : no prior knowledge for what an appropriate p_t and u_t are.
- p_t : many choices of probability paths $p_1(x) \approx q$
- u_t : not a closed form that generates pt

→ **construct** p_t **and** u_t

3.1 Constructing p_t, u_t from Conditional Probability Paths and Vector Fields

A simple way to construct a target probability path p_t

= making it a mixture of simpler probability paths that are only defined per sample

- **Conditional Probability** $p_1(x|x_1)$
  - given particular data sample x1
  - t=0: $p_0(x|x_1) = p(x)$
  - t=1: $p_1(x|x_1) = \mathcal{N}(x|x_1, \sigma^2 I)$
- **Marginal Probability Path** $p_t(x) = \int p_t(x|x_1)q(x_1)dx_1$
  - marginalizing the conditional probability paths over $q(x_1)$
  - t=1: $p_1(x) = \int p_1(x|x_1)q(x_1)dx_1 \approx q(x)$

3.1 Constructing p_t, u_t from Conditional Probability Paths and Vector Fields

A simple way to construct a target probability path p_t

= making it a mixture of simpler probability paths that are only defined per sample

- **Conditional Vector Field $u_t(x|x_1)$**
  - a conditional vector field $u_t(\cdot|x_1) : \mathbb{R}^d \to \mathbb{R}^d$ that generates $p_t(\cdot|x_1)$
- **Marginal Vector Field**
  - marginalizing the conditional vector fields over $q(x_1)$
  - $$u_t(x) = \int u_t(x|x_1) \frac{p_t(x|x_1)q(x_1)}{p_t(x)} dx_1$$

3.1 Constructing p_t, u_t from Conditional Probability Paths and Vector Fields

Observation #1

The marginal vector field generates the marginal probability path

$$u_t(x) = \int u_t(x|x_1)\frac{p_t(x|x_1)q(x_1)}{p_t(x)}dx_1 \rightarrow \; p_t(x) = \int p_t(x|x_1)q(x_1)dx_1$$

Theorem 1.

Given vector fields $u_t(x|x_1)$ that generate conditional probability paths $p_t(x|x_1)$,

for any distribution $q(x_1)$, **the marginal vector field u_t generates the marginal probability path p_t**, i.e., u_t and p_t satisfy the *continuity equation*

Theorem 1: $u_t(x|x_1)$이 $p_t(x|x_1)$를 생성하면, Eq. 8의 $u_t(x)$가 Eq. 6의 $p_t(x)$를 생성한다.

증명 핵심 (연속방정식 검증):

$$\frac{d}{dt}p_t(x) = \int \frac{d}{dt}p_t(x|x_1)\cdot q(x_1)dx_1 \quad \text{(Leibniz)}$$

$$= -\int \mathrm{div}(u_t(x|x_1)p_t(x|x_1))q(x_1)dx_1 \quad \text{(조건부 연속방정식)}$$

$$= -\mathrm{div}\left(\int u_t(x|x_1)p_t(x|x_1)q(x_1)dx_1\right) \quad \text{(div과 적분 교환)}$$

$$= -\mathrm{div}(u_t(x)p_t(x)) \quad \text{(Eq. 8의 정의)}$$

- 연속방정식 성립 → u_t가 p_t를 생성
- Eq. 8의 가중평균 정의가 이 증명이 성립하도록 역설계된 것

3.2 Conditional Flow Matching

Issue

- **still intractable** to compute u_t
  - intractable integrals in the definitions of the marginal probability path and vector fields
  - intractable to compute an unbiased estimator of the original FM objective

→ **propose CFM objective** instead of FM objective : tractable, closed

Conditional Flow Matching (CFM) objective

- where $t \sim \mathcal{U}[0,1]$, $x_1 \sim q(x_1)$, and now $x \sim p_t(x|x_1)$

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p_t(x|x_1)} \left\| v_t(x) - u_t(x|x_1) \right\|^2$$

3.2 Conditional Flow Matching

	FM (Eq. 5)	CFM (Eq. 9)
위치 샘플링	$x \sim p_t(x)$ (주변)	$x_1 \sim q(x_1), x \sim p_t(x\|x_1)$ (조건부)
회귀 타겟	$u_t(x)$ (주변 VF)	$u_t(x\|x_1)$ (조건부 VF)
계산 가능?	불가능	**가능**

Observation #2

The FM and CFM objectives have identical gradients w.r.t. θ.

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,p_t(x)}\|v_t(x) - u_t(x)\|^2$$

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p_t(x|x_1)}\left\|v_t(x) - u_t(x|x_1)\right\|^2 \rightarrow \nabla_\theta \mathcal{L}_{FM}(\theta) = \nabla_\theta \mathcal{L}_{CFM}(\theta)$$

Theorem 2.

Assuming that pt(x) > 0 for all x ∈ R d and t ∈ [0, 1], then, up to a constant independent of θ, $\mathcal{L}_{CFM}$ and $\mathcal{L}_{FM}$ are equal. Hence, $\nabla_\theta \mathcal{L}_{FM}(\theta) = \nabla_\theta \mathcal{L}_{CFM}(\theta)$

- 증명: L2 노름의 이중선형성으로 전개 후, θ 의존 항이 동일함을 보임
- **CFM 최적화 = FM 최적화 → marginal pt, ut 없이 CNF 훈련이 가능하다**

증명 전략: L2 노름의 이중선형성으로 전개 후, θ 의존 항이 동일함을 보임

Step 1 — 전개:

$$\|v_t - u_t\|^2 = \|v_t\|^2 - 2\langle v_t, u_t\rangle + \|u_t\|^2$$

$$\|v_t - u_t(\cdot|x_1)\|^2 = \|v_t\|^2 - 2\langle v_t, u_t(\cdot|x_1)\rangle + \|u_t(\cdot|x_1)\|^2$$

- $\|u_t\|^2, \|u_t(\cdot|x_1)\|^2$는 θ에 무관 → 그래디언트에서 소멸

Step 2 — $\|v_t\|^2$ 항이 같음:

$$\mathbb{E}_{p_t(x)}\|v_t(x)\|^2 = \int \|v_t(x)\|^2 p_t(x)dx \overset{\text{Eq.6}}{=} \iint \|v_t(x)\|^2 p_t(x|x_1)q(x_1)dx_1dx$$

$$\overset{\text{Fubini}}{=} \mathbb{E}_{q(x_1),p_t(x|x_1)}\|v_t(x)\|^2$$

Step 3 — 내적 항이 같음 (핵심):

$$\mathbb{E}_{p_t(x)}\langle v_t, u_t\rangle = \int \left\langle v_t(x), \frac{\int u_t(x|x_1)p_t(x|x_1)q(x_1)dx_1}{p_t(x)}\right\rangle p_t(x)dx$$

- $p_t(x)$가 약분됨
- 내적의 선형성으로 적분을 안으로 이동
- Fubini로 적분 순서 교환

$$= \mathbb{E}_{q(x_1),p_t(x|x_1)}\langle v_t(x), u_t(x|x_1)\rangle$$

**결론: θ 의존 항 모두 동일 → $\nabla_\theta \mathcal{L}_{FM} = \nabla_\theta \mathcal{L}_{CFM}$

4. Conditional Probability Paths and Vector Fields

Gaussian conditional probability paths $p_t(x|x_1) = \mathcal{N}(x \mid \mu_t(x_1), \sigma_t(x_1)^2 I)$

- t=0: $\mu_0(x_1) = 0 \;\; \sigma_0(x_1) = 1$
- t=1: $\mu_1(x_1) = x_1 \;\; \sigma_1(x_1) = \sigma_{\min}$ → small so that $p_1(x|x_1)$ is a concentrated Gaussian distribution centered at x1

Corresponding flow

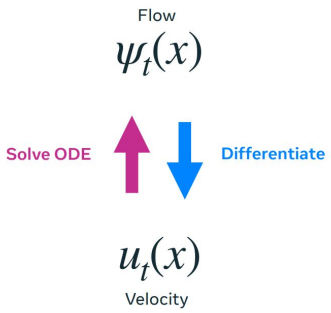
$$\psi_t(x) = \sigma_t(x_1)x + \mu_t(x_1)$$

- standard Gaussian x . affine transformation $\psi_t(x)$
- Push-forward $[\psi_t]_* p(x) = p_t(x|x_1)$

Vector Field

$$\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)|x_1)$$

- flow provides a vector field that generates the conditional probability path



4. Conditional Probability Paths and Vector Fields

Reparameterized CFM Loss

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p_t(x|x_1)} \left\| v_t(x) - u_t(x|x_1) \right\|^2$$

Reparameterizing $p_t(x|x_1)$ in terms of just x_0 and plugging $\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)|x_1)$

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p(x_0)} \left\| v_t(\psi_t(x_0)) - \frac{d}{dt}\psi_t(x_0) \right\|^2$$

Theorem 3. → **make compute possible by defining a closed form vector field** $u_t(x|x_1)$

Let $p_t(x|x_1)$ be a Gaussian probability path, and ψ_t its corresponding flow map.

Then, the **unique vector field** that defines ψ_t has the form $u_t(x|x_1) = \frac{\sigma_t'(x_1)}{\sigma_t(x_1)}(x - \mu_t(x_1)) + \mu_t'(x_1)$

Consequently, $u_t(x|x_1)$ generates the Gaussian path $p_t(x|x_1)$

4. Conditional Probability Paths and Vector Fields

Example I : Diffusion conditional Vector Fields

- VE (Variance Exploding) Path

$$p_t(x|x_1) = N(x|x_1, \sigma_{1-t}^2 I) \quad \Rightarrow \quad \mu_t = x_1,\ \sigma_t = \sigma_{1-t}$$

$$u_t(x|x_1) = -\frac{\sigma'_{1-t}}{\sigma_{1-t}}(x - x_1) \quad \text{(Eq. 17)}$$

- VP (Variance Preserving) Path

$$p_t(x|x_1) = N(x \mid \alpha_{1-t}x_1, (1-\alpha_{1-t}^2)I), \quad \alpha_t = e^{-\frac{1}{2}T(t)},\ T(t) = \int_0^t \beta(s)ds$$

$$u_t(x|x_1) = -\frac{T'(1-t)}{2}\left[\frac{e^{-T(1-t)}x - e^{-\frac{1}{2}T(1-t)}x_1}{1 - e^{-T(1-t)}}\right] \quad \text{(Eq. 19)}$$

- **Diffusion Path**는 유한 시간에 진정한 노이즈 분포에 도달하지 못함
- → **FM**은 직접 설정하므로 이 문제 없음

4. Conditional Probability Paths and Vector Fields

Example II: Optimal Transport conditional Vector Fields

- Path : 평균과 표준편차가 시간에 대해 선형

$$\mu_t(x_1) = tx_1, \quad \sigma_t(x_1) = 1 - (1 - \sigma_{\min})t$$

- **Conditional flow** that corresponds to $u_t(x|x_1)$

$$\psi_t(x) = (1 - (1 - \sigma_{\min})t)x + tx_1$$

- **CFM Loss**

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p(x_0)} \left\| v_t(\psi_t(x_0)) - \Big(x_1 - (1 - \sigma_{\min})x_0\Big) \right\|^2$$

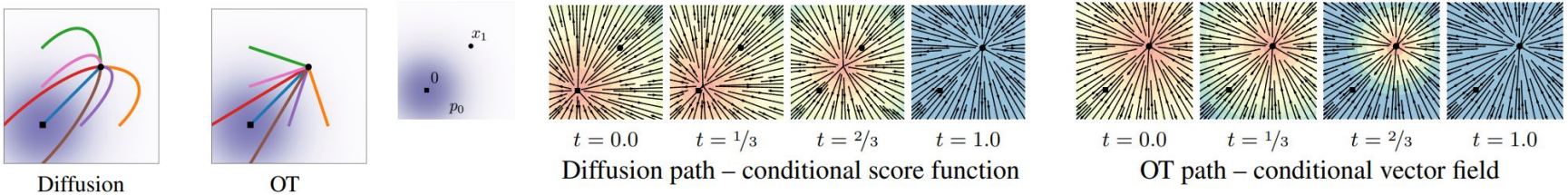


Figure 3: Diffusion and OT trajectories.

5. Related Work

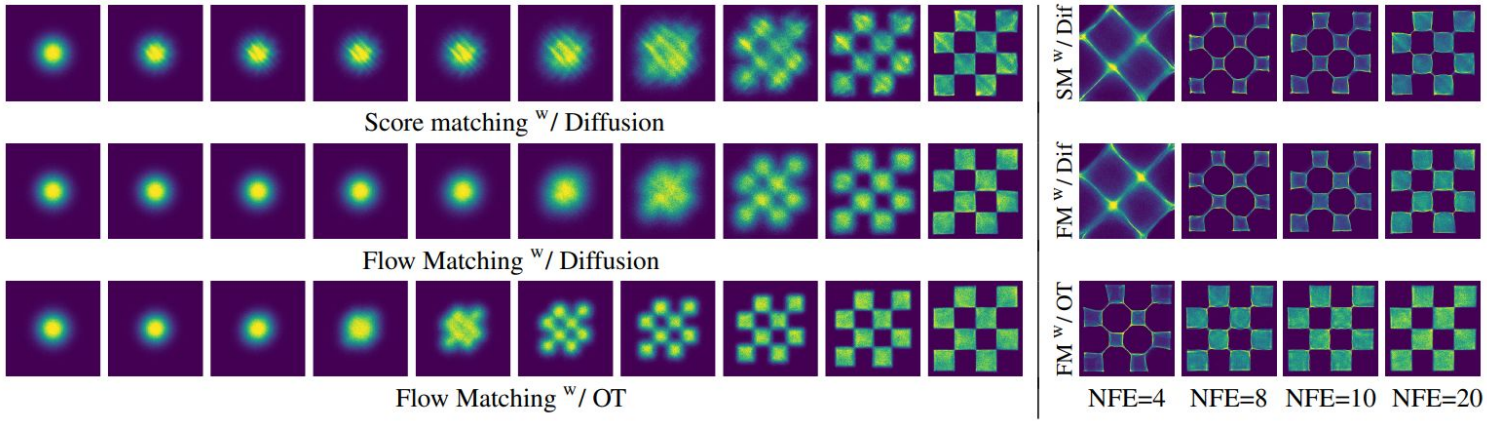


Figure 4: (*left*) Trajectories of CNFs trained with different objectives on 2D checkerboard data. The OT path introduces the checkerboard pattern much earlier, while FM results in more stable training. (*right*) FM with OT results in more efficient sampling, solved using the midpoint scheme.

- Optimal Transport displacement interpolation (직선으로 가는 길)이 직선 궤적과 일정 속도를 보장
- Diffusion 경로 대비 더 단순한 회귀 타겟

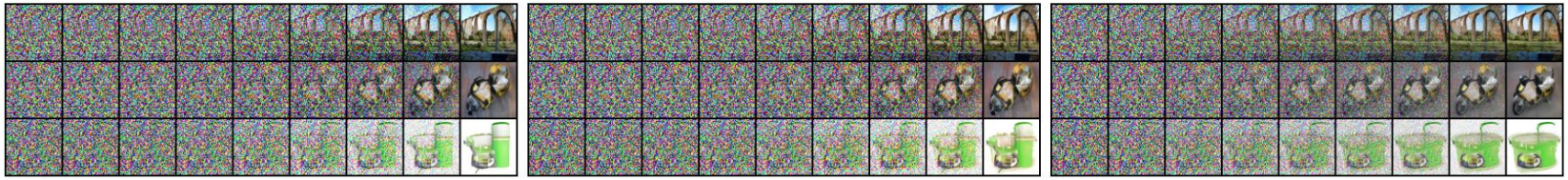
6. Experiments

Model	CIFAR-10			ImageNet 32×32			ImageNet 64×64		
	NLL↓	FID↓	NFE↓	NLL↓	FID↓	NFE↓	NLL↓	FID↓	NFE↓
Ablations									
DDPM	3.12	7.48	274	3.54	6.99	262	3.32	17.36	264
Score Matching	3.16	19.94	242	3.56	5.68	178	3.40	19.74	441
ScoreFlow	3.09	20.78	428	3.55	14.14	195	3.36	24.95	601
Ours									
FM w/ Diffusion	3.10	8.06	183	3.54	6.37	193	3.33	16.88	187
FM w/ OT	**2.99**	**6.35**	**142**	**3.53**	**5.02**	**122**	**3.31**	**14.45**	**138**

Model	ImageNet 128×128	
	NLL↓	FID↓
MGAN (Hoang et al., 2018)	–	58.9
PacGAN2 (Lin et al., 2018)	–	57.5
Logo-GAN-AE (Sage et al., 2018)	–	50.9
Self-cond. GAN (Lučić et al., 2019)	–	41.7
Uncond. BigGAN (Lučić et al., 2019)	–	25.3
PGMGAN (Armandpour et al., 2021)	–	21.7
FM w/ OT	**2.90**	**20.9**

Table 1: Likelihood (BPD), quality of generated samples (FID), and evaluation time (NFE) for the same model trained with different methods.

- FM-OT가 세 지표 모두에서 최고 성능



Score Matching w/ Diffusion Flow Matching w/ Diffusion Flow Matching w/ OT

Figure 6: Sample paths from the same initial noise with models trained on ImageNet 64×64. The OT path reduces noise roughly linearly, while diffusion paths visibly remove noise only towards the end of the path. Note also the differences between the generated images.

6.1 Density Modeling and Sample Quality on ImageNetvb

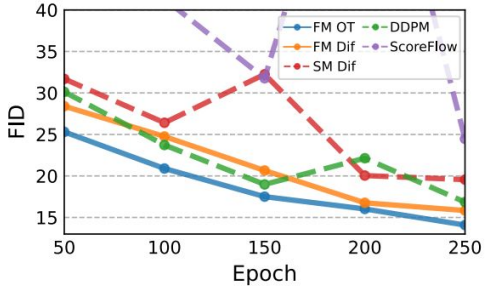
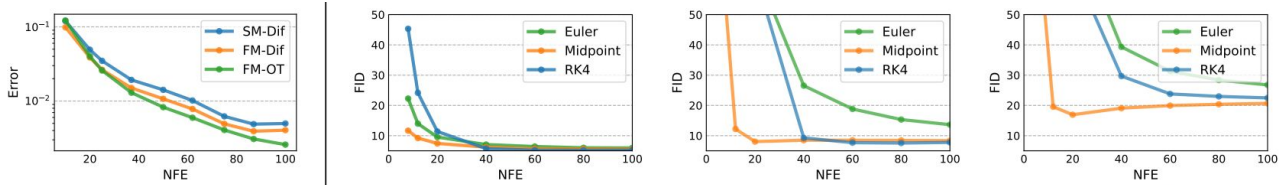


Figure 5: Image quality during training, ImageNet 64×64.

Model	FID↓	IS↑	PSNR↑	SSIM↑
Reference	1.9	240.8	–	–
Regression	15.2	121.1	**27.9**	**0.801**
SR3 (Saharia et al., 2022)	5.2	180.1	26.4	0.762
FM ^{w}/ OT	**3.4**	**200.8**	24.7	0.747

Table 2: Image super-resolution on the ImageNet validation set.

6.2 Sampling Efficiency



Error of ODE solution | Flow matching ^{w}/ OT | Flow matching ^{w}/ Diffusion | Score matching ^{w}/ Diffusion

Figure 7: Flow Matching, especially when using OT paths, allows us to use fewer evaluations for sampling while retaining similar numerical error (left) and sample quality (right). Results are shown for models trained on ImageNet 32×32, and numerical errors are for the midpoint scheme.

7. Conclusion

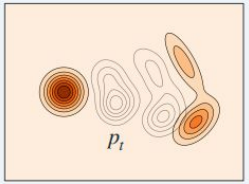
Flow Matching

Defined by any $p_{t|1}$ we know how to generate

- arbitrary p, q and coupling
- $p_{t|1}$
- u_t, X_0, X_1
- u_t, cond-OT/linear $\Rightarrow$ performance, efficiency

CTMP

$X_t \sim p_t$



Data

Conditional paths

Velocity Param

Gaussian affine paths

Process

Diffusion models (deterministic)

Defined by forward process (data → noise)

- $p = \mathcal{N}$, independent coupling
- $\mathcal{N}(\alpha_t x_1, \sigma_t^2 I)$ singularity
- ϵ, X_1, v

Flow, Diffusion, CTMC

ThankYou